

Data Science Capstone Project Report

SpaceX Falcon 9 First-Stage Landing Prediction

Completed presentation slides in PDF format

IBM Data Science Capstone | Dataset period: 2010-2020

GitHub URL of the Project

<https://github.com/TristinWen/spacex-falcon9-landing-prediction>

Public grading repository

Completed notebooks | Dash source and CSV | Dash screenshots | Folium HTML map | Final PDF

Executive Summary

Methods

SpaceX API, Wikipedia web scraping, data wrangling, EDA visualization, SQL, Folium, Plotly Dash, and four classification models.

90

Falcon 9 launches

87.8%

prepared-label success
share

Key results

KSC LC-39A contributes 41.7% of successful launches. All four classifiers achieve 83.3% test accuracy; decision tree leads cross-validation at 92.9%.

Introduction: Project Background and Problem

Background

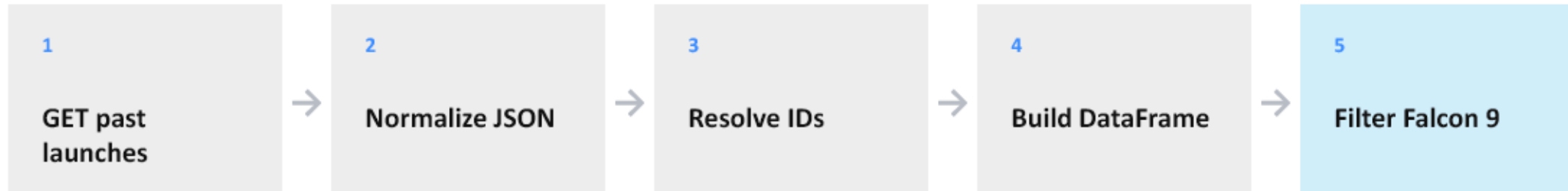
Falcon 9 first stages can be recovered and reused. Recovery can lower launch cost and affect competitive bid estimates.

Problem statement

Identify the mission characteristics associated with successful landing and predict whether the first stage will land successfully.

Target: Class = 1 for successful landing; Class = 0 for unsuccessful landing.

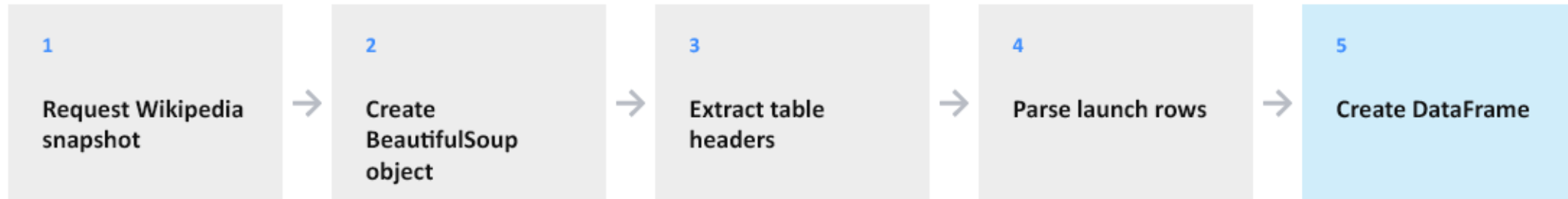
Data Collection - SpaceX API



Output: 90 Falcon 9 launches and 17 analytical fields.

Key fields: payload mass, orbit, launch site, landing outcome, reuse history, grid fins, legs, block, coordinates.

Data Collection - Web Scraping



Output: 121 historical launch rows with 11 fields.

Fields include flight number, date, time, booster, site, payload, mass, orbit, customer, launch outcome, and landing outcome.

Data Wrangling Methodology

Cleaning and processing steps

- 01 Calculate missing-value percentages
- 02 Replace missing PayloadMass with the mean
- 03 Retain LandingPad nulls as meaningful no-pad cases
- 04 Count launch sites and orbit occurrences
- 05 Map unsuccessful outcomes to Class = 0
- 06 Map remaining outcomes to Class = 1

Result

90 rows
0 missing PayloadMass
87.8% success share

EDA with Data Visualization: Methods and Chart Purpose

Scatter plots

Test relationships among flight number, payload, site, orbit, and landing class

Bar chart

Compare average landing success across orbit types

Line chart

Show the yearly landing-success trend

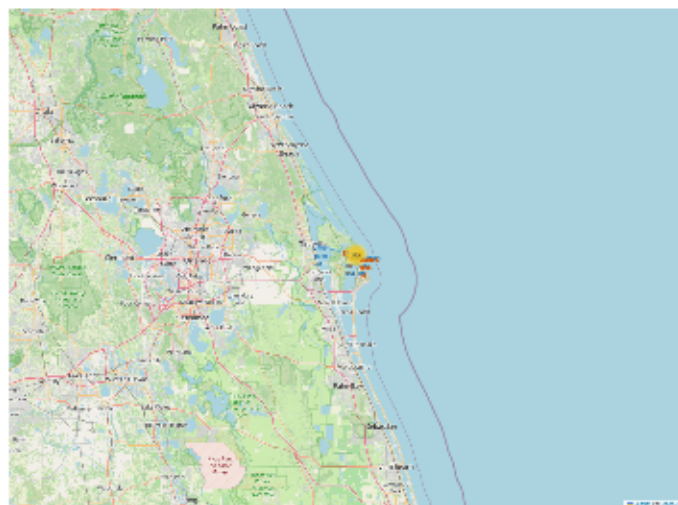
Result slides follow in Rubric 1.11 order.

EDA with SQL: Query Summary

- Unique launch sites and CCA site records
- Total and average payload mass
- First successful ground-pad landing date
- Boosters with successful drone-ship landings
- Mission success and failure totals
- Maximum-payload booster versions
- Monthly failure records and landing-outcome ranking

SQL result slides follow in Rubric 1.12 order.

Interactive Visual Analytics: Folium and Plotly Dash

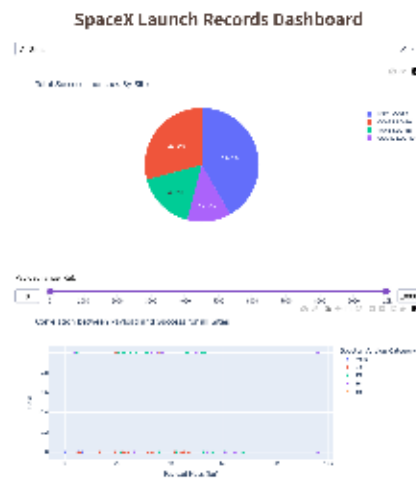


Folium

Launch-site markers, outcome clusters, and proximity distances.

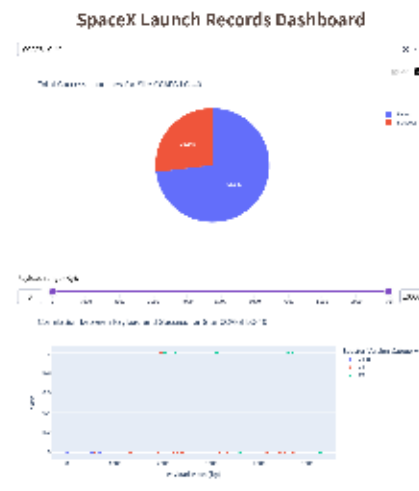
Folium: https://github.com/TristinWen/spacex-falcon9-landing-prediction/blob/main/notebooks/lab_jupyter_launch_site_location_completed.ipynb

Dash: <https://github.com/TristinWen/spacex-falcon9-landing-prediction/blob/main/dashboard/spacex-dash-app.py>

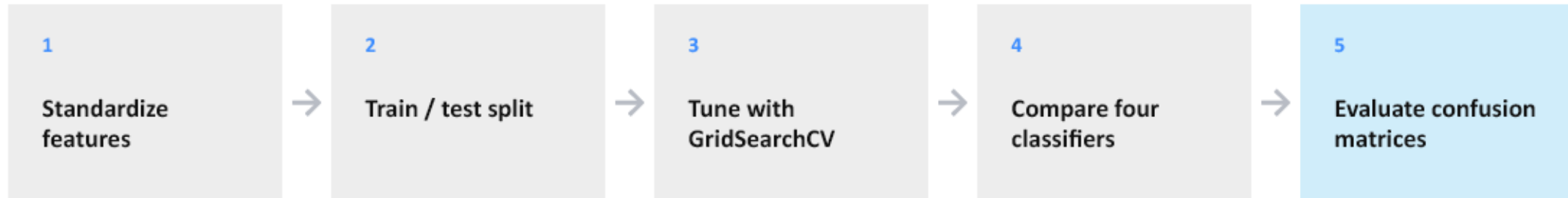


Plotly Dash

Site dropdown, success pie charts, payload slider, and payload-versus-outcome scatter plots.



Predictive Analysis Methodology



Models: Logistic Regression | Support Vector Machine | Decision Tree | K-Nearest Neighbors

Selection uses cross-validation performance; final evaluation uses the held-out test set and confusion matrices.

EDA Result: Flight Number vs. Launch Site



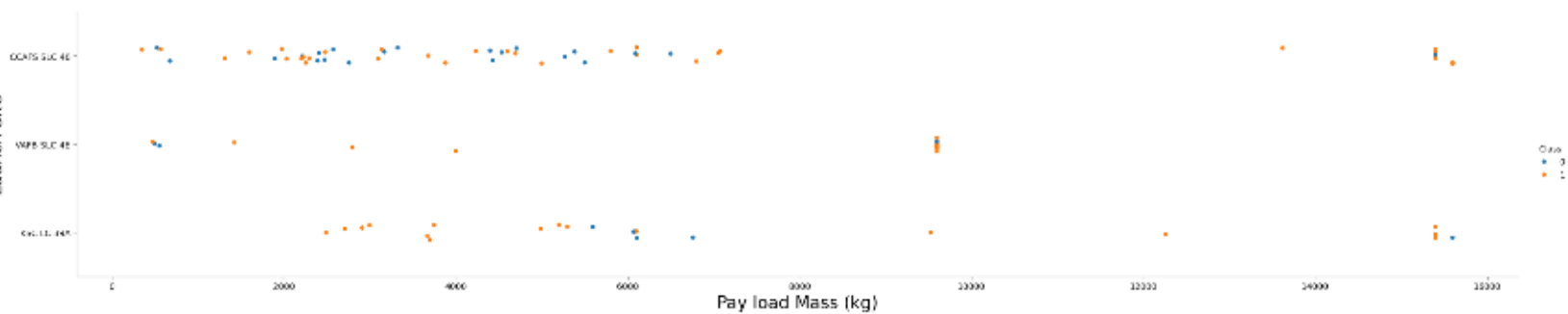
Purpose

Assess whether landing success changes with launch experience at each site.

Finding

Later flights show more successful outcomes across launch sites.

EDA Result: Payload Mass vs. Launch Site



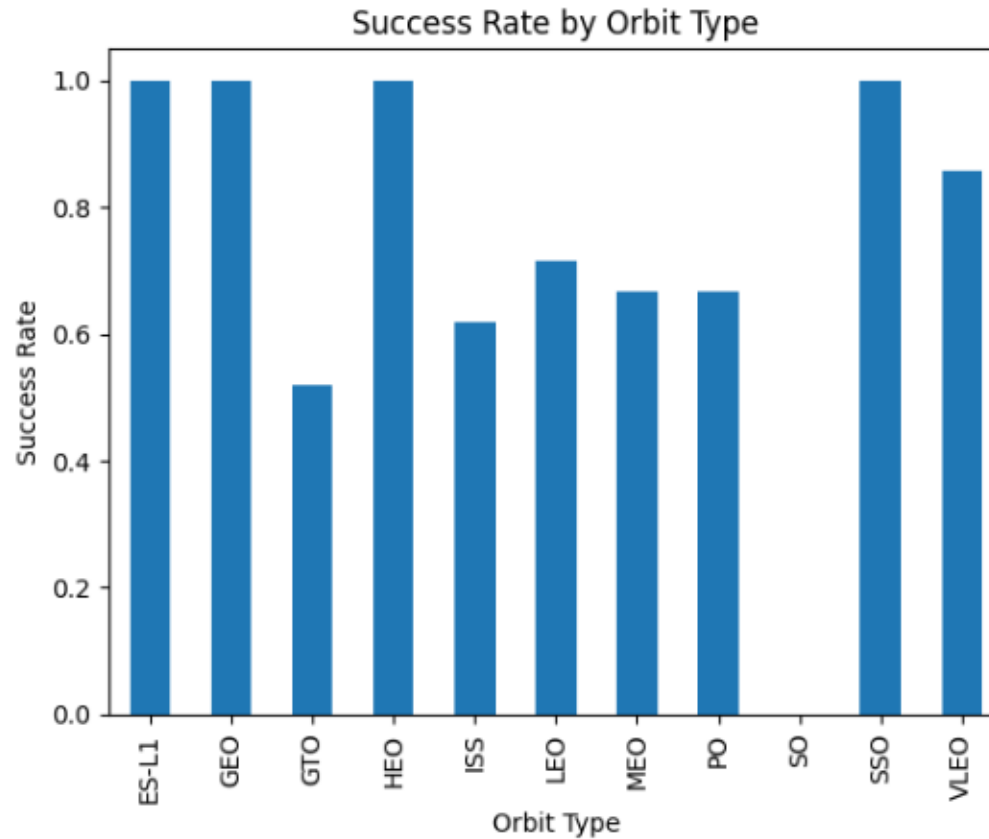
Purpose

Compare payload ranges and landing outcomes by launch site.

Finding

Sites operate across different payload ranges and outcome patterns.

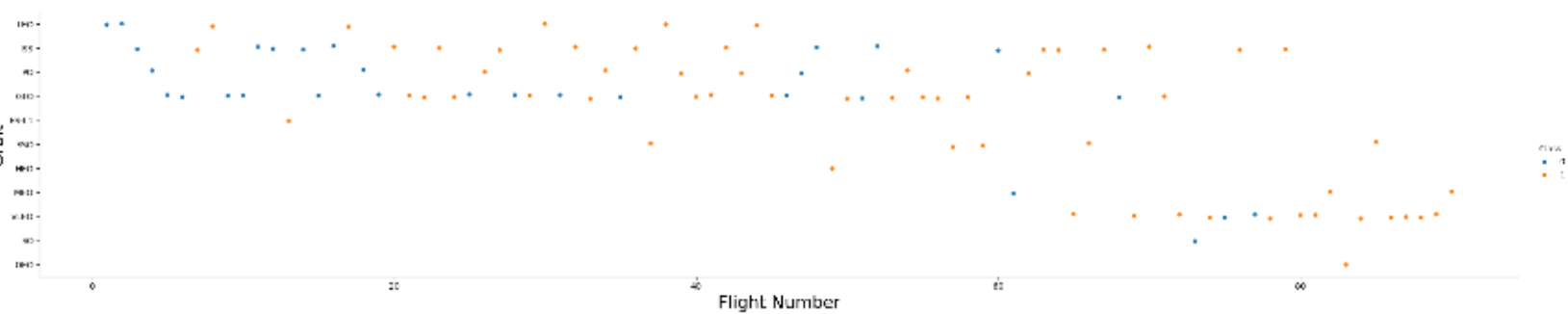
EDA Result: Success Rate by Orbit Type



The bar chart compares mean Class by orbit.

ES-L1, GEO, HEO, and SSO show perfect success in this sample, while GTO is materially lower.

EDA Result: Flight Number vs. Orbit Type



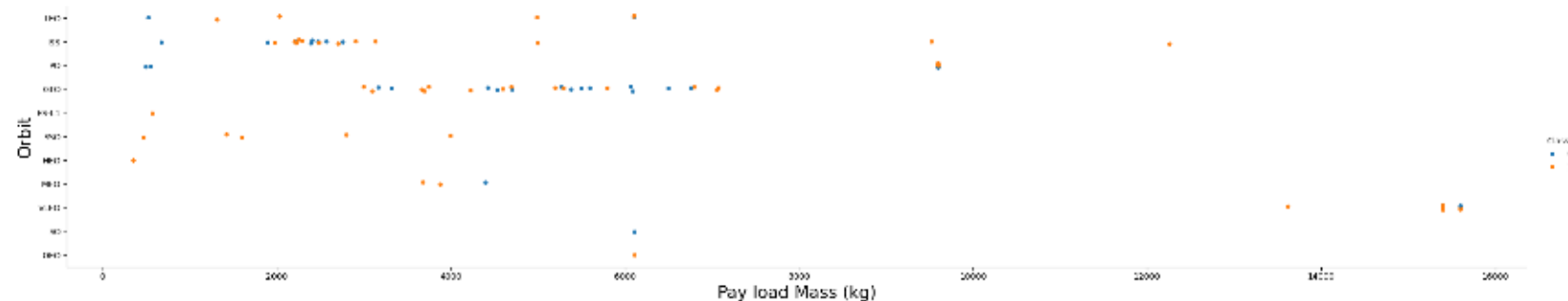
Purpose

Show how orbit usage and landing success evolve across flight sequence.

Finding

Several orbit profiles improve as flight number increases.

EDA Result: Payload Mass vs. Orbit Type



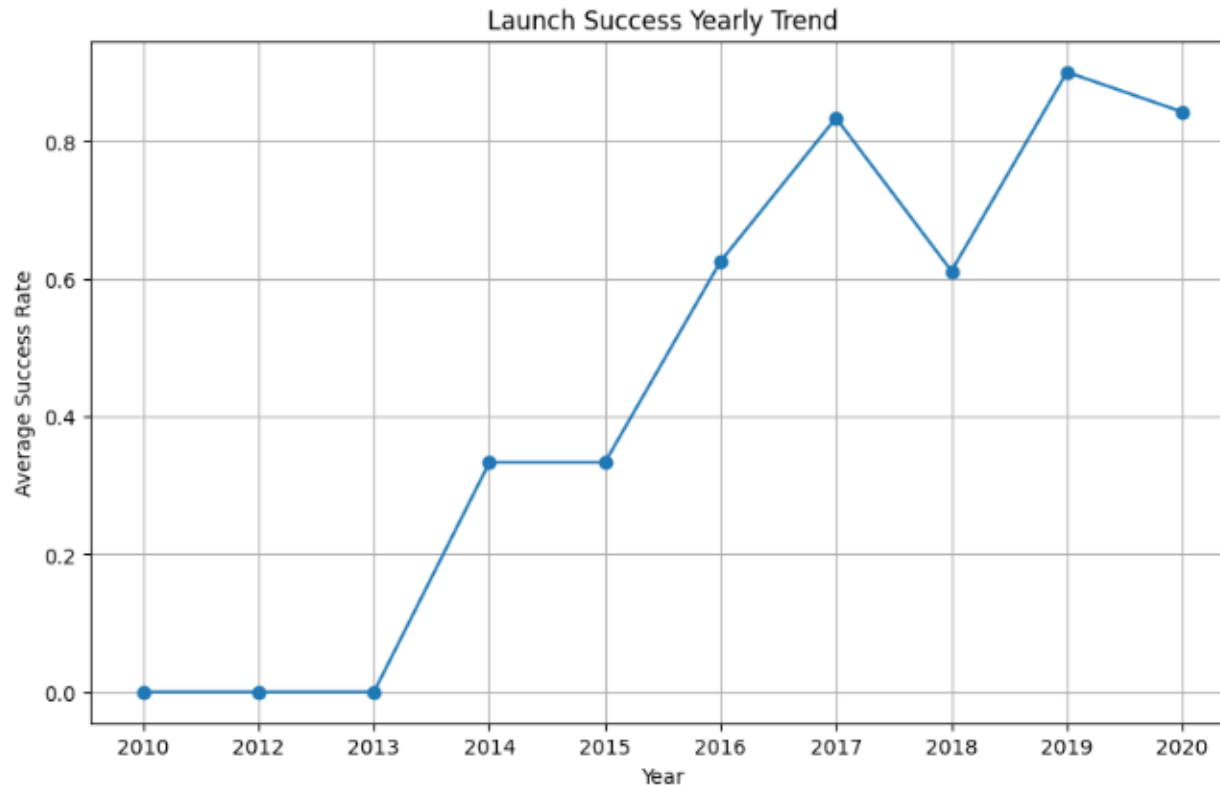
Purpose

Compare mission payload ranges and outcomes across orbit types.

Finding

Payload-outcome relationships differ by orbit and mission profile.

EDA Result: Yearly Launch Success Trend



Finding

Average landing success rises from zero in the early years to above 80% by 2019-2020, with a temporary dip in 2018.

EDA with SQL Results: Launch Sites and Payloads

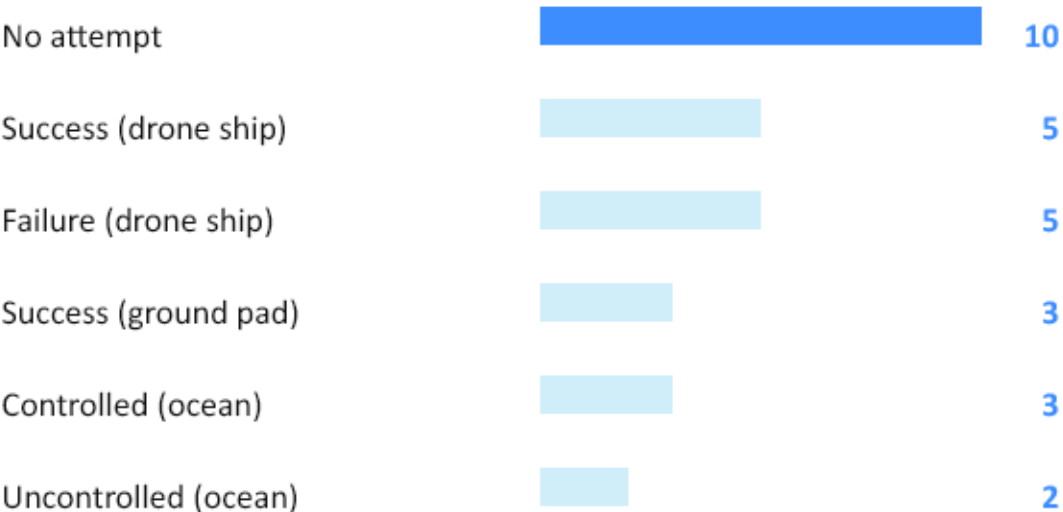
Unique launch sites	CCAFS LC-40; CCAFS SLC-40; KSC LC-39A; VAFB SLC-4E
CCA launch records	First five requested records returned
NASA CRS total payload	45,596 kg
F9 v1.1 average payload	2,928.4 kg
Maximum-payload boosters	12 Falcon 9 Block 5 booster records

EDA with SQL Results: Success Rates and Rankings

Mission outcomes

Success: 98
Success (trailing-space variant): 1
Success (payload unclear): 1
Failure in flight: 1

Landing outcome ranking, 2010-2017



Successful drone-ship boosters at 4,000-6,000 kg: B1022, B1026, B1021.2, B1031.2

EDA with SQL Results: Time Analysis

First successful ground-pad landing

22 December 2015

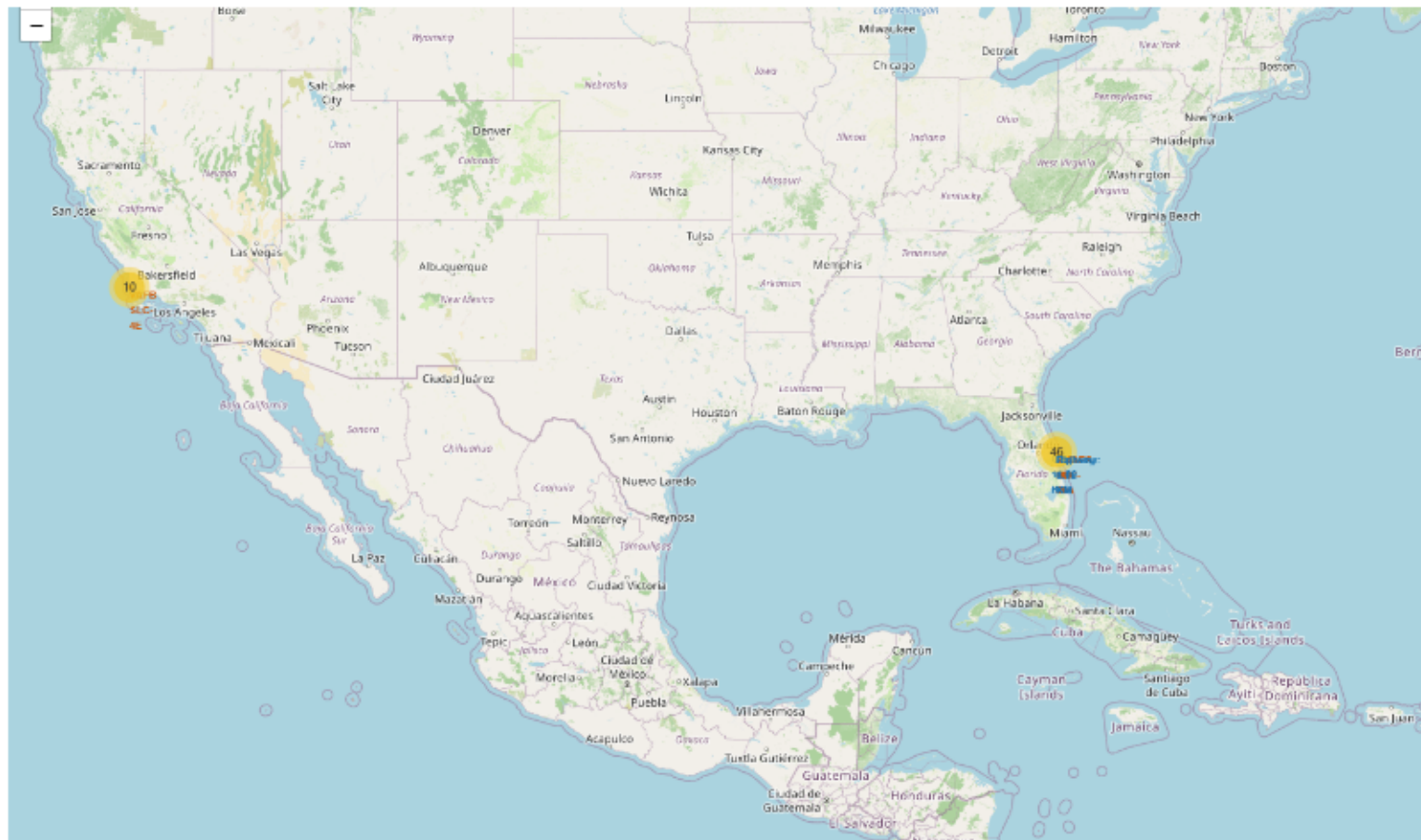
2015 drone-ship failures

January - F9 v1.1 B1012 - CCAFS LC-40

April - F9 v1.1 B1015 - CCAFS LC-40

The ordered SQL analysis covers launch sites, payloads, success outcomes, rankings, dates, and months.

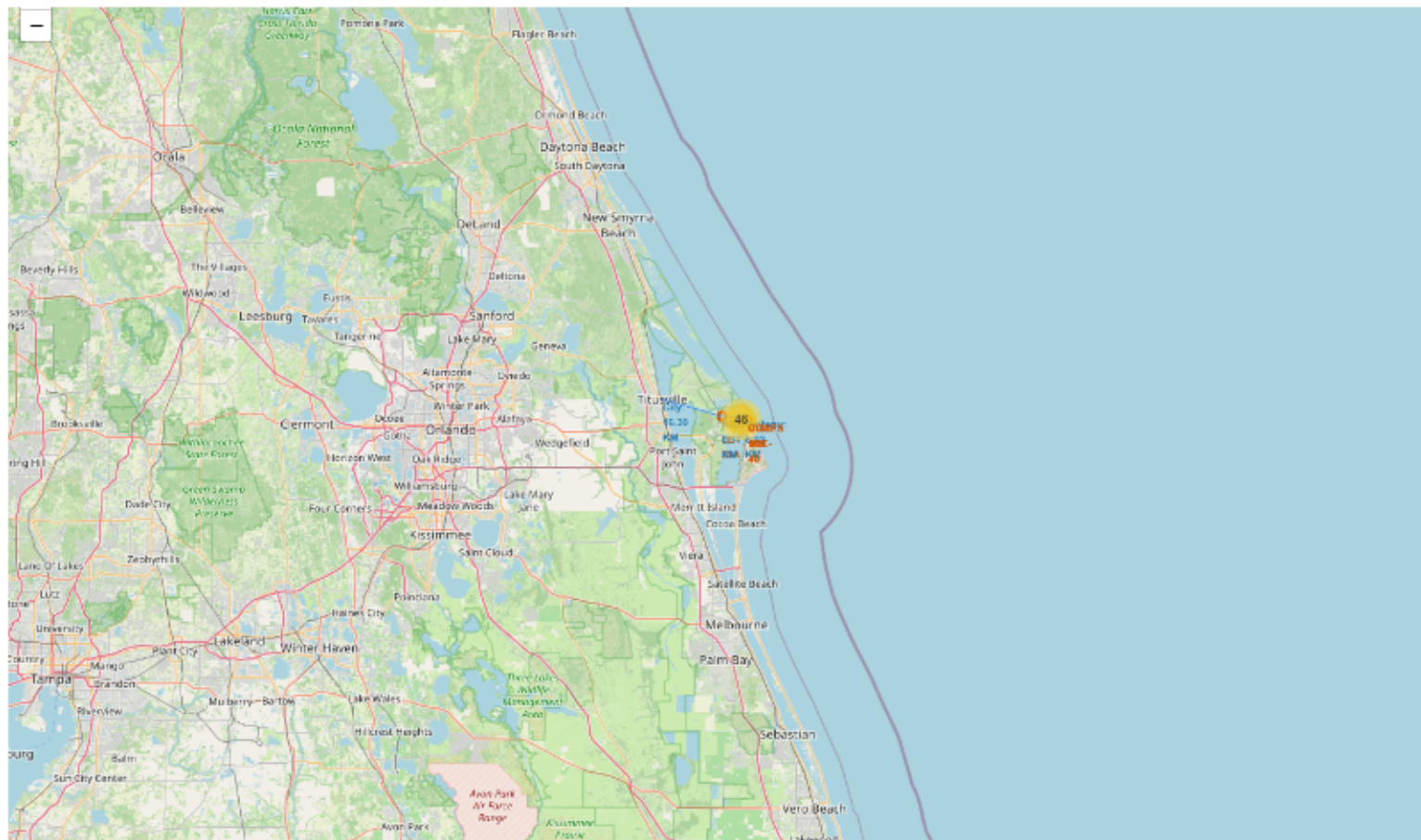
Folium Map Result: Launch Site Markers



Map evidence

- Four launch-site labels
- Coastal locations in Florida and California
- Marker clusters summarize overlapping launch records

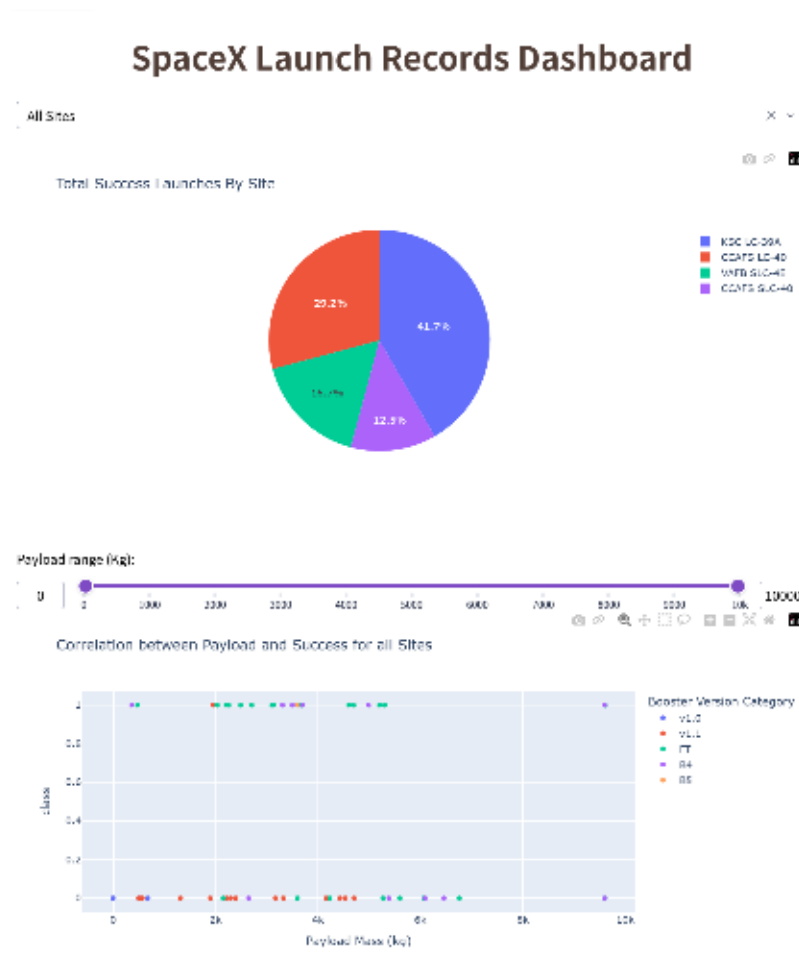
Folium Map Result: Launch Records and Proximity Analysis



Completed evidence

- Green/red success and failure markers
- Distance to coastline
- Lines to city, railway, and highway points
- Launch sites are close to coastlines and transport infrastructure

Plotly Dash Result: Success Pie Chart

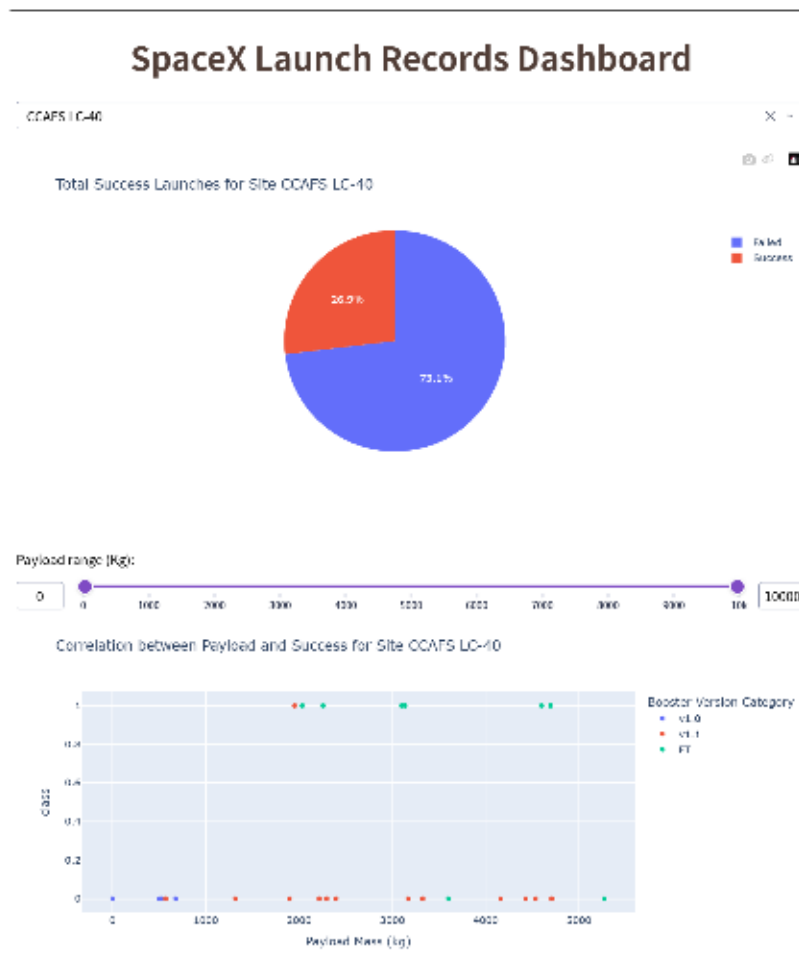


41.7%

KSC LC-39A has the largest share of successful launches.

Required controls visible: site dropdown and payload range slider.

Plotly Dash Result: Payload vs. Launch Outcome Scatter Plot



CCAFS LC-40

26.9% success
73.1% failure

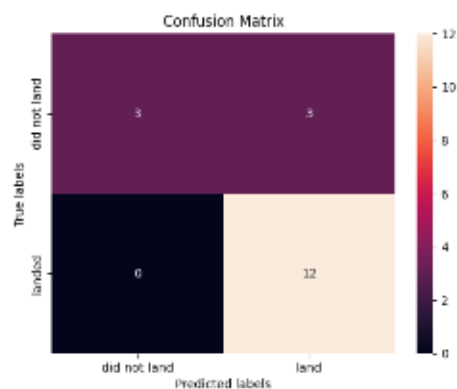
The site filter updates both the success/failure pie and payload-versus-outcome scatter plot.

Predictive Analysis: Multiple Models and Evaluation Results

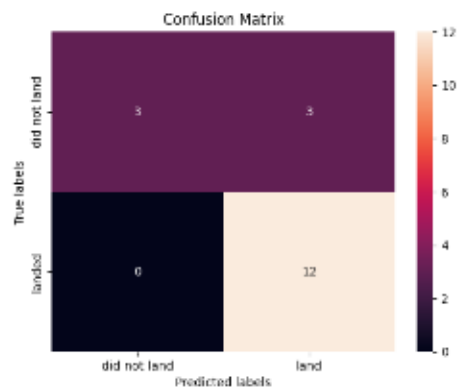
Model	CV accuracy	Test accuracy
Logistic Regression	84.6%	83.3%
Support Vector Machine	84.8%	83.3%
Decision Tree	92.9%	83.3%
K-Nearest Neighbors	84.8%	83.3%

All models were tuned with GridSearchCV; the test set contains 18 launches.

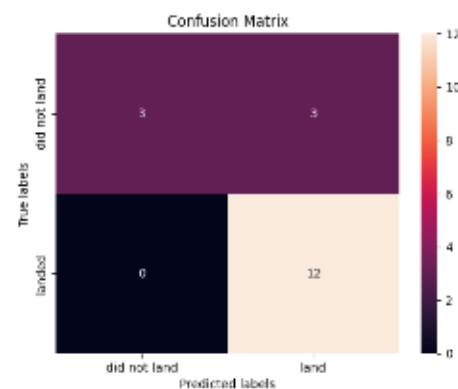
Predictive Analysis: Confusion Matrices



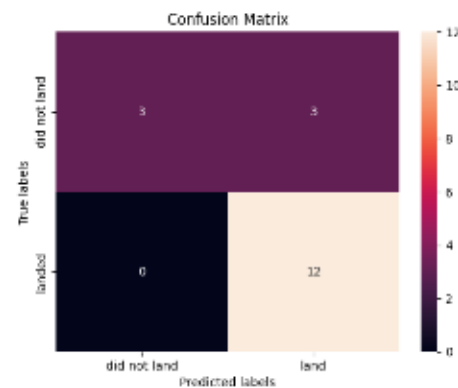
Logistic Regression



Decision Tree



Support Vector Machine



K-Nearest Neighbors

Best Model Explanation and Final Conclusion

Best cross-validation model

Decision Tree

92.9% validation accuracy

83.3% test accuracy

Why this choice

The decision tree captures nonlinear interactions among site, orbit, payload, booster block, reuse history, grid fins, and legs.

Creative insight

Use calibrated landing probabilities, not only binary labels, to estimate expected launch cost and bid risk. Expand the dataset and use repeated cross-validation before operational deployment.

Conclusion and Future Work

What the analysis shows

- Landing success improved substantially over time.
- Launch site, orbit, payload, and flight maturity interact.
- KSC LC-39A contributes the largest share of successful launches.
- Decision Tree provides the strongest cross-validation result.

Recommended next steps

Monitor model drift as new launches arrive.

Add weather, pad-condition, and detailed trajectory variables.

Use calibrated probabilities for cost and bid-risk estimates.

Validate the model on newer, out-of-time launches.

The model is decision support for engineers—not a replacement for mission review.

Creative Presentation Elements

A visual story rather than a notebook dump

- 01 Consistent visual hierarchy and color system across the full project journey
- 02 Flow diagrams explain collection and modeling steps at a glance
- 03 Maps and dashboard screenshots demonstrate interactive work
- 04 Purpose-and-finding panels turn every chart into an interpreted result
- 05 A presentation roadmap helps peer reviewers locate every evidence group

Innovative Insights Beyond the Required Tasks

Interaction insight

Payload, orbit, launch site, and flight maturity should be interpreted together. Their effects are not independent, so a single payload threshold cannot explain landing success across all missions.

Decision insight

Calibrated landing probabilities could connect model output to expected recovery cost and bid risk. This is more actionable than a binary prediction alone, but it requires more data and out-of-time validation.

Project Journey and Evidence Map

01	GitHub repository	Slide 2	09	SQL Results	Slide 18-20
02	Completed PDF	Slide 1-31	10	Folium Results	Slide 21-22
03	Executive Summary	Slide 3	11	Plotly Dash Results	Slide 23-24
04	Introduction	Slide 4	12	Predictive Results	Slide 25-27
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06	EDA & Interactive Methods	Slide 8-10	14	Creativity	Slide 29
07	Predictive Methodology	Slide 11	15	Innovative Insights	Slide 30
08	EDA Visualization Results	Slide 12-17			